

Assignment CA-2

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Acknowledgement

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Declaration

We, Riya (Registration Number: 12316877), Shivang Chaurasia (Registration Number: 12316885), and Himanshu Shekhar (Registration Number: 12318302), hereby declare that the project titled **“ParkEase:- Cost Effective Parking Solution”** has been carried out by us and is an original work.

The project was completed during the period from **April, 2026 to May, 2026** as part of the academic requirements for the award of the degree of **Bachelor of Technology (B. Tech) in Computer Science and Engineering** at **Lovely Professional University (LPU)**.

We confirm that this work has not been submitted earlier, either wholly or partially, to any other university or institution for the award of any degree or diploma. All references, resources, and contributions used during the course of this project have been properly acknowledged.

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Introduction

The rapid expansion of urban populations and vehicle ownership has led to massive congestion, inefficient space utilization, and daily frustrations for commuters seeking reliable parking. However, traditional parking systems and generic reservation platforms often struggle to cater specifically to modern mobility needs, frequently lacking real-time availability, transparent pricing, and secure payment integrations. To address these limitations, **ParkEase** has been developed as a sophisticated, **SaaS-grade** smart mobility operating system designed to facilitate an organized, real-time parking ecosystem for both drivers and property hosts.

ParkEase is built using a robust full-stack architecture powered by **Laravel 11** and **MongoDB**, utilizing **Blade templates**, **Bootstrap 5**, and **vanilla JavaScript** to deliver a premium, responsive **glassmorphism aesthetic**. The platform integrates modern features such as **secure passwordless identity management via Clerk**, a production-ready UPI-first payment flow via **Razorpay**, advanced geospatial search (Haversine filtering), and automated **PDF invoice generation**. Focusing on scalability, operational clarity, and **seamless UI/UX**, the system converts fragmented urban parking inventory into an intelligent, searchable, and highly reliable digital mobility network.

System Modules

The **ParkEase** platform is divided into multiple functional modules to ensure scalability, maintainability, and efficient system operation. Each module performs a specific role within the overall architecture of the system.

1. User Authentication and Authorization Module

This module manages secure onboarding, login, session states, and multi-role access control. Powered by **Clerk.js**, it implements robust **password less** identity management, seamlessly differentiating between drivers looking for parking spaces and property hosts managing inventory. It also includes an automated **KYC** (Know Your Customer) gating flow, ensuring that all parking hosts are fully verified before their spaces go live on the platform.

2. Geospatial Discovery and Interactive Search Module

This module handles the core search functionality, enabling users to effortlessly find available parking spaces.

Leveraging **MongoDB** for high-performance data querying, it utilizes the Haversine distance formula to filter and display parking lots based on exact **GPS** coordinates or user-entered **pincodes**. It also integrates an interactive visual map with real-time indicators to display parking slot availability.

3. Booking Lifecycle and Session Management Module

This module manages the end-to-end flow of parking reservations. It organizes user activities into a highly structured, intelligent dashboard featuring categorized session states (Active, Upcoming, Past, and Cancelled). It implements live countdown timers for active parking sessions and ensures real-time synchronization between the user's booking status and the host's inventory availability.

4. Financial Transactions and Smart Refund Module

This module handles all monetary exchanges within the platform securely and reliably. It integrates the **Razor pay** SDK to facilitate a production-ready, UPI-first payment flow complete with cryptographic signature verification. Additionally, it enforces intelligent, time-based cancellation logic that automatically calculates and processes refunds (e.g., 100%, 50%, or 0% windows) based on how close to the start time a booking is **canceled**.

5. Automated Invoicing and Digital Ticketing Module

This module focuses on post-transaction record-keeping and validation. Using **Barryvdh DomPDF**, it automatically generates professional, downloadable PDF parking tickets and payment receipts the moment a booking is confirmed. It seamlessly embeds dynamically generated QR validation tokens into the digital tickets to allow hosts to easily authenticate drivers upon arrival.

6. Host Inventory and Operations Dashboard Module

This module provides parking space owners with a dedicated control panel to monitor and manage their physical assets. It includes utilities to list new parking spaces, define hourly pricing structures, monitor live occupancy rates, and review incoming bookings or cancellations, empowering hosts to maximize the utilization of their real estate.

Tools and Technologies Used

The development of the ParkEase platform involved the use of modern tools and technologies to ensure scalability, security, high performance, and a premium user experience.

Frontend Technologies

• Blade Template & Vanilla JavaScript – Used for server-side rendering of dynamic pages combined with lightweight client-side interactivity to ensure fast page loads and seamless user interactions. • Vite – Employed as the frontend build tool and asset bundler for fast compiling, live reloading, and efficient asset delivery. • Bootstrap 5 & Custom CSS3 – Used to build a clean, consistent, and responsive user interface following a premium, modern glassmorphism aesthetic. • LottieFiles – Integrated to render high-quality, lightweight micro-animations that enhance the dynamic and premium feel of the platform.

Backend Technologies

• Laravel 11 (PHP 8.4+) – Used as the core MVC framework to handle robust routing, controllers, business logic, complex booking lifecycle management, and backend security features.

Database

• MongoDB Atlas – Used as a NoSQL database via the mongodb/laravel-mongodb integration, enabling efficient, schema-less storage for complex structures like user bookings, host inventories, and high-performance geospatial (Haversine) search capabilities.

Authentication & Security

• Clerk.js – Utilized as a comprehensive Identity-as-a-Service solution to manage secure onboarding, passwordless authentication, user sessions, and multi-role (User/Host) access control.

Payment Processing

• Razorpay SDK – Integrated to facilitate secure, production-ready financial transactions, offering a UPI-first payment flow, cryptographic signature verification, and automated dynamic refunds.

Document Generation

• Barryvdh DomPDF – Employed on the backend to dynamically generate customized, downloadable PDF parking tickets and invoices embedded with QR validation tokens.

Version Control & Deployment

• Git & GitHub – Used for version control, code management, and collaborative teamwork among developers. • Render & Docker – Utilized for containerized production hosting, scalable infrastructure orchestration, and continuous integration/continuous deployment (CI/CD) of the web application.

Development Tools

• Visual Studio Code – Primary IDE utilized for writing, formatting, and debugging application code.

GitHub Repository Link

https://github.com/ShivangChaurasia/ParkEase-Cost-Effective-Parking_System

Deployed Link

<https://parkease-3as5.onrender.com/>

Instagram Post Link

<https://www.instagram.com/reel/DYucHLzy1Ry/?igsh=djVtdTkzaHNmeWxp>

